

Omesh Sharma

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Profile

Senior Software Engineer with 7.5+ years of experience building distributed systems, high-performance backend platforms, and enterprise-scale applications using C/C++, Python, and cloud-native technologies. Experienced across Oracle BRM, messaging infrastructure, Document AI, quantitative systems, and VR/graphics platforms. Strong expertise in system design, multithreading, performance optimization, secure communication, debugging, and high-availability architectures.

Professional Summary

Core Strengths: Strong problem-solving and decision-making skills with an ownership mindset focused on designing scalable, secure, and efficient systems. Experienced in architecting products from scratch, troubleshooting complex production issues, optimizing distributed systems, and delivering enterprise-grade backend solutions

Experience & Domains: Enterprise Applications, Telecom & Communication Systems, OSS/BSS (BRM, CRM, ERP), Email Infrastructure, Distributed Backend Systems, Desktop Applications, VR/AR & Graphics Systems, High-Frequency Trading Platforms, Quantitative Systems, and Document AI platforms

AI & Specialized Areas: Document AI, OCR, OpenCV, Vision-based Processing, NLP, Feature Engineering, ML Concepts, RAG Workflows, LLM Integrations, Agent-based Systems, Quantitative Development, TA-Lib, Cryptocurrency and Trading Systems

Technical Proficiency: C/C++, Qt Framework, Python, Java, JavaScript, REST APIs, FastAPI, Distributed Systems, Backend Engineering, Parallel Processing, Multithreading, High Availability, and Performance Optimization

Frameworks & Platforms: FastAPI, Django, Streamlit, Docker, Kubernetes, Helm, OCI, Azure Cloud, Jenkins, Azure DevOps, CI/CD, Prometheus, Grafana, Elasticsearch, Kafka, Microservices, and Cloud-Native Architectures

Databases & Storage: Oracle DB, MySQL, Cassandra, Elasticsearch, JSON, Samba NAS, Azure Blob Storage, Object Storage, and Distributed Storage Systems

Networking & Security: SSL/TLS, TCP/IP, SMTP, DNS, SPF/DKIM, Authentication Systems, JWT, Secure Communication Protocols, REST/SOAP APIs, NAS Systems, and Secure Distributed Communication

Core Engineering Skills: System Architecture, Distributed Systems, OOP, STL, Concurrency, Multithreading, Automation, Debugging, Root Cause Analysis, System Reliability, and Performance Engineering

Tools & DevOps: Git, GitHub, GitLab, Jenkins, SonarCloud, Docker, Kubernetes, Helm, Prometheus, Grafana, GDB, Valgrind, tcpdump, Wireshark, Selenium, Apitrace, and Autolt

Project Management & Collaboration: Agile Development, Sprint Planning, SDLC, Technical Ownership, Architecture Discussions, Mentoring, Cross-Team Collaboration, Customer-Facing Issue Resolution, and Technical Documentation

Experience

Oracle, Senior Member of Technical Staff

Bengaluru, Karnataka, India

Leading development across Oracle Billing and Revenue Management (BRM) and Oracle Messaging Server, focusing on distributed systems, secure communication, high availability, platform modernization, and enterprise-scale backend infrastructure.

Apr 2022 – present

- Led development and optimization of Oracle Messaging Server and BRM systems handling secure communication, routing, delivery, and distributed transaction processing.
- Designed Active-Active and Active-Passive HA architectures improving reliability, failover handling, and zero-downtime capabilities across enterprise deployments.
- Enabled TLS 1.3, FIPS compliance, Oracle Linux 9 migration, and security modernization across distributed services and legacy environments.
- Resolved complex Sev1/Sev2 customer production issues involving SSL/TLS, networking, memory leaks, socket handling, distributed communication, and dependency conflicts.

- Worked closely with enterprise customers during go-live, migration, upgrade, and cloud transformation activities, ensuring deployment stability and production readiness.
- Helped customers troubleshoot scalability bottlenecks, HA failover behavior, mail routing inconsistencies, authentication failures, and distributed system integration issues.
- Contributed to BRM business logic enhancements involving billing workflows, rating, taxation, usage processing, transaction handling, and backend service integrations.
- Implemented REST APIs and backend automation services in Python for BRM integrations, operational tooling, and infrastructure automation.
- Contributed to Java modernization (JDK 8 to 21), platform compatibility improvements, and cloud-native deployment readiness.
- Improved observability, debugging, profiling, and RCA workflows across large-scale messaging and billing infrastructure.
- Worked extensively on SMTP, TCP/IP, SSL/TLS, IPv6, SPF/DKIM, authentication systems, OCI, Kubernetes, Docker, Cassandra, Elasticsearch, and Oracle DB.
- Assisted customers in configuring secure mail forwarding, relaying, anti-spam policies, secure communication flows, and enterprise mail security standards.
- Enhanced messaging server communication layers, retry mechanisms, timeout handling, queue management, and secure socket interactions for high-scale deployments.
- Built AI-assisted internal knowledge base and troubleshooting workflows to improve engineering productivity, faster issue resolution, and reusable operational guidance.
- Collaborated across platform, infrastructure, security, and integration teams to deliver reliable enterprise-scale distributed solutions.

dox2U, Software Engineer

Gurugram, Haryana, India
June 2020 – Apr 2022

Built a cloud-based Document AI and OCR platform from scratch focused on intelligent document management, distributed processing, and automation.

- Architected a distributed Document AI platform for scanned documents, PDFs, invoices, and searchable document processing.
- Developed high-performance OCR and parser engines using C++, Python, FastAPI, OpenCV, Tesseract OCR, and ImageMagick.
- Built intelligent document classification, preprocessing, skew correction, orientation detection, and multilingual OCR pipelines.
- Designed distributed PDF processing systems enabling parallel execution and searchable PDF generation.
- Integrated Kafka, Samba NAS, Azure Blob Storage, and Elasticsearch for scalable processing and real-time search.
- Containerized microservices with Docker and implemented monitoring, CI/CD, Sonar-Cloud quality gates, and Azure DevOps workflows.

Hooklabs, Quantitative Developer

Delhi, India
June 2019 – June 2020

Designed and developed a quantitative trading and backtesting platform for automated cryptocurrency trading and market analysis.

- Architected and developed a quantitative trading and backtesting engine using C++, Qt, Python, and TA-Lib.
- Built automated trading workflows integrating Binance APIs and real-time market data processing.
- Implemented genetic algorithms, optimization techniques, and feature engineering for market strategy evaluation.
- Processed large-scale historical datasets using concurrent and parallel processing models.
- Developed analytical modules for profitability, risk analysis, and quantitative trading metrics.

VizExperts, Software Engineer - Graphics & VR/AR Systems

Delhi, India
Jan 2019 – June 2019

Worked on enterprise VR/AR and graphics applications using OpenSceneGraph, OpenGL, and Qt-based desktop systems.

- Developed save/reload and replay frameworks for VR sessions using OpenSceneGraph, Qt, JSON serialization, and rendering pipelines.

- Designed Qt-based integration bridges connecting CATIA, Navisworks, and Aveva applications with VR systems.
- Built API tracing, rendering analysis, and automation frameworks using Python and Autolt.
- Worked extensively on OpenGL/OSG-based desktop applications, rendering workflows, and CAD integrations.

Education

B.tech Chandigarh Engineering College, Computer Science and Engineering

Mohali, Punjab, India

- Strong foundation in Data Structures, Operating Systems, Networking, Algorithms, DBMS, and Computer Architecture.

July 2015 – Dec 2018

Achievements

- Designed and optimized enterprise-scale distributed systems across telecom, messaging, OCR, trading, and graphics domains.
- Strong expertise in system-level debugging, performance optimization, secure communication, and high-availability architectures.
- Delivered production-critical fixes and architecture improvements for large-scale Oracle communication and billing platforms.